Math is Explaining and Sharing

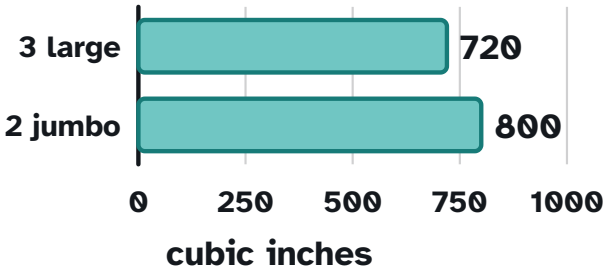
Use volume to compare boxes and defend your choice with numbers.

Name _____ **Date** _____

Tonight's spark

A corner store sells cereal at one price: three large boxes or two jumbo boxes. A large box is 8 by 3 by 10 inches. A jumbo box is 10 by 4 by 10 inches. Which deal holds more cereal?

Make a guess first: three large boxes or two jumbo boxes?



Our class model: multiply the three sides

Try it

Work it out on paper. Write one answer on each line.

1. **Volume is measured in which kind of unit?**

   A. Square units **B.** Cubic units **C.** Units of length

   My answer _____

2. **What is the volume of a box that is 2 inches long, 3 inches wide, and 4 inches tall?**

   A. 24 cubic inches **B.** 9 cubic inches **C.** 24 square inches

   My answer _____

3. **Which number sentence finds the volume of a box 5 in by 2 in by 6 in?**

   A. 5 + 2 + 6 **B.** 5 × 2 **C.** 5 × 2 × 6 **D.** (5 × 2) + 6

   My answer _____

4 **A box is 6 inches long, 5 inches wide, and 2 inches tall. What is its volume?**

A. 13 cubic inches **B.** 30 cubic inches **C.** 600 cubic inches **D.** 60 cubic inches

My answer _____

5 **Box A holds 90 cubic inches. Box B holds 120 cubic inches. Which statement is true?**

A. Box A holds more, because 90 is smaller

B. Box B holds more, because 120 is greater than 90 **C.** They hold the same amount

My answer _____

Talk about it

Which box holds more, and what numbers prove it?

Home mission

Find a cereal box and a snack box. Measure each side in inches with a ruler. Multiply to find each volume. Can your student say which box holds more and finish the sentence with "because"?

If we get stuck, ask

- What are the three measurements of this box?
- Which total is bigger — and how do you know?
- Can you say your answer starting with "because"?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/1-4/homework.html

How did it feel? (optional)

Tricky Getting there Got it